Jupiter

The gas giant - fifth planet from the Sun

Overview

Jupiter is the fifth planet from the Sun and the largest planet in the Solar System. It is a gas giant made mostly of hydrogen and helium, and its mass is more than two and a half times that of all the other planets combined. It is named after the king of the Roman gods.

Jupiter has no solid surface. Its visible face consists of colourful cloud bands of ammonia crystals, separated into light 'zones' and darker 'belts', driven by powerful winds.

Key facts

- **Average distance from the Sun:** 5.2 AU (about 778 million km)
- **Diameter:** 139,820 km (about 11 times Earth)
- **Length of day:** 9 hours 56 minutes (shortest of all planets)
- **Length of year:** 11.86 Earth years
- **Surface gravity:** 24.79 m/s squared (about 2.5 times Earth, at cloud tops)
- **Average temperature:** about -110 C (at cloud tops)
- **Number of moons:** more than 90 known
- **Axial tilt:** 3.1 degrees

Structure and atmosphere

By volume, Jupiter's atmosphere is about 90% hydrogen and 10% helium. Deep inside, the enormous pressure turns hydrogen into a liquid that conducts electricity like a metal. Electric currents in this 'metallic hydrogen' layer generate Jupiter's magnetic field, the strongest of any planet.

Jupiter also has a faint ring system made of dust, discovered by the Voyager 1 spacecraft in 1979.

The Great Red Spot

The **Great Red Spot** is a giant storm, larger than Earth, located in Jupiter's southern hemisphere. It has been observed continuously since at least the 1830s and possibly since the 17th century. Wind speeds at its edges exceed 400 km/h. Observations show the storm has been gradually shrinking over the past century.

Moons

Jupiter's four largest moons are called the **Galilean moons**, because Galileo Galilei discovered them in 1610:

Io - the most volcanically active body in the Solar System.

Europa - has an icy crust covering a global ocean of liquid water, making it a key target in the search for life.

Ganymede - the largest moon in the Solar System; it is larger than the planet Mercury and has its own magnetic field.

Callisto - one of the most heavily cratered objects in the Solar System.

Formation and influence

Jupiter was probably the first planet to form in the Solar System. Because of its huge mass, its gravity has strongly influenced the other planets and small bodies. It shapes the asteroid belt, creating gaps where few asteroids are found, and it shares its orbit with thousands of **Trojan asteroids**, which gather in two groups ahead of and behind the planet. NASA's **Lucy** mission, launched in October 2021, is on its way to visit several of them.

Jupiter radiates more heat into space than it receives from the Sun. This extra energy comes from the planet slowly contracting and releasing heat left over from its formation.

Magnetosphere and radiation

Jupiter's magnetic field creates a huge magnetosphere - the largest structure belonging to any planet in the Solar System. It traps charged particles in intense radiation belts that can damage spacecraft electronics, so missions such as Juno use shielded vaults to protect their instruments. Jupiter also has powerful, permanent auroras at its poles.

Comet impact of 1994

In July 1994 more than 20 fragments of **Comet Shoemaker-Levy 9** crashed into Jupiter over several days. It was the first collision between two Solar System bodies ever observed directly, and the impacts left dark scars in Jupiter's atmosphere, some larger than Earth, that remained visible for months.

Observing Jupiter from Earth

Jupiter is usually the fourth-brightest object in the night sky, after the Sun, the Moon and Venus. Even a pair of binoculars shows the four Galilean moons as small points of light that change position from night to night, and a small telescope reveals the cloud belts.

Exploration history

- **Pioneer 10** (NASA, 1973) - first spacecraft to fly past Jupiter.
- **Voyager 1 and 2** (NASA, 1979) - discovered the rings and volcanoes on Io.
- **Galileo** (NASA, in orbit 1995-2003) - first spacecraft to orbit Jupiter; dropped a probe into its atmosphere.
- **Juno** (NASA, arrived 4 July 2016) - studies Jupiter's interior, gravity and magnetic field from a polar orbit.
- **JUICE** (ESA, launched April 2023) - will study Ganymede, Callisto and Europa; arrival planned for 2031.
- **Europa Clipper** (NASA, launched October 2024) - will study Europa's ocean; arrival planned for 2030.

Did you know?

- Jupiter has the shortest day of any planet - under 10 hours.
- More than 1,300 Earths could fit inside Jupiter.

- Ganymede, a moon of Jupiter, is bigger than the planet Mercury.